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From forest to farmland: Agroecosystem evolution and sustainable development strategies

Summary

As the global population grows, the ecological problems caused by agricultural expansion have attracted much attention. This article focuses on habitat change from forests to farms, as well as the impact of natural and human factors on agricultural ecosystems, in order to promote agricultural maturity and sustainable development.

First, based on the case of forest-land conversion in the Brazilian legal Amazon (BLA), we built an agri-food web model to simulate an ecosystem that has just been converted from forest to farmland. Second, to measure the impact of chemicals, We consider seasonality and combine **Lotka-Volterra** models and create the **Se-LV model** to establish the differential equations of various groups. We then mapped how the numbers of various groups changed over time (Figures 5 and 6), and the results showed that the use of chemicals destabilizes ecosystems.

Then, we reintroduced snakes and short-tailed cats into the ecosystem, combined with the Se-LV model to establish differential equations for various groups, and then we used MATLAB to map the number of various groups over time (Figure 7). In addition, we pretreated the population with **Min-Max**, and finally calculated that the standard deviation **std (X)** of the population is less than 1, indicating that the ecosystem stability is enhanced after species regression.

Next, we removed the herbicides from the ecosystem and simulated the process using a **cellular automaton model**. Then, we counted the proportion of each population according to the size of each color region in the animated image and performed **spearman correlation analysis** on the data. **Lasso regression** was used to report the stability of the ecosystem. The results showed that herbicides are beneficial to crop growth in the short term but affect ecological stability and biodiversity in the long term

Then, we introduced bats and frogs to build a beneficial species food web model, and then combined the Se-LV model to establish differential equations for various groups, plot the number of various groups over time (Figure 10), and finally calculate the population standard deviation. It is found that frogs are more effective in controlling pest populations, promoting biological population growth, and enhancing ecosystem stability.

Then, we analyzed the impact of organic farming, using **cellular automata models** and **kinetic equations** to simulate organic farming. In addition, we established a cost-benefit formula using the idea of **linear programming** to calculate the maximum production benefit. The results showed that organic farming has significant advantages in various aspects while ensuring efficiency, which is conducive to long-term sustainable development.

Finally, we evaluated and extended the model's strengths and weaknesses. We also analyzed the sensitivity of each parameter. In addition, we prepared a letter to a farmer who was exploring organic farming methods.

Keywords: Lotka-Volterra model; cellular automata model; lasso regression; spearman correlation analysis

Contents

1 Introduction **3**
1.1 Problem Background 3
1.2 Restatement of the Problem 3
1.3 Our Work 4
2 Assumptions and Justifications **4**
3 Notations **5**
4 Agroecosystem food web model **5**
4.1 Components of a food web model 5
4.2 Building a food web model 6
4.3 Dynamic changes in each component 8
5 Species Regression Models **9**
5.1 The reture of snake 10
5.2 The reture of short-tailed cat 10
5.2.1 Changes in the number of species 10
5.2.2 Stability of ecosystems 11
6 Herbicide removal models **12**
6.1 Cellular Automata Model 12
6.2 Ecosystem stability report 14
7 Beneficial species food web model **15**
7.1 Introduction of bats 15
7.2 Introduction of frogs 16
7.3 Comparison of beneficial species 17
7.3.1 Number of species 17
7.3.2 Stability of ecosystems 17
8 Organic farming models **18**
8.1 The concept of organic farming 18
8.2 Cellular Automata Model 18
8.3 Long-term sustainability and cost-effectiveness 20
9 Sensitivity Analysis **21**
10 Model Evaluation and Further Discussion **22**
10.1 Strengths 22
10.2 Weaknesses 22

10.3 Further Discussion 22
11 Conclusion...23
12 References 23

1 Introduction

1.1 Problem Background

As the world's population grew, agricultural expansion became an important means of meeting human food needs. Not only did it result in habitat loss, but after deforestation, the fertility of the soil began to decline, and the originally fertile land gradually became impoverished. At the same time, due to the loss of control of natural predators, pests began to invade crops, leaving farmers to rely on chemicals to protect crops. Although this approach solved the problem of pests in the short term, it destroyed the ecological balance of the soil and further exacerbated land degradation.

In this context, agroecosystems have gradually formed a new kind of food web. Bats and birds play important roles in agroecosystems, for example, bats provide ecosystem services to crops by preying on pests12[1], while birds affect crop yields to some extent. However, this human-driven ecosystem requires a long period of adjustment and adaptation to reach a relatively stable state.

Therefore, how to find a balance between agricultural expansion and ecological protection has become an urgent problem to be solved. Therefore, we establish a food web model from forest to farmland to promote agricultural maturity and sustainable development.

1.2 Restatement of the Problem

Changing habitats from forests to farmland is a complex process that involves multiple issues. To determine the evolution of this ecosystem and changes in agricultural decision-making, we need to complete the following questions:

Problem 1: Create a food web model for an ecosystem that has recently transitioned from forests to agriculture, and study the cyclical, seasonal, and chemical effects of agriculture on that ecosystem.

Problem 2: Predict the impact of the re-emergence of two species on the agroecosystem.

Problem 3: Report the stability of ecosystems after herbicides from the perspective of producers and consumers.

Problem 4: Predict the impact of bat inclusion in food web models on the overall stability of the ecosystem, and identify similar beneficial species to compare their impacts.

Problem 5: Analyze the implications of a farmer considering organic farming methods and

assess the impact of this method on ecosystems and their components.
Problem 6: Write a letter to farmers who are exploring organic farming practices.

1.3 Our Work

So we were asked to do the following.

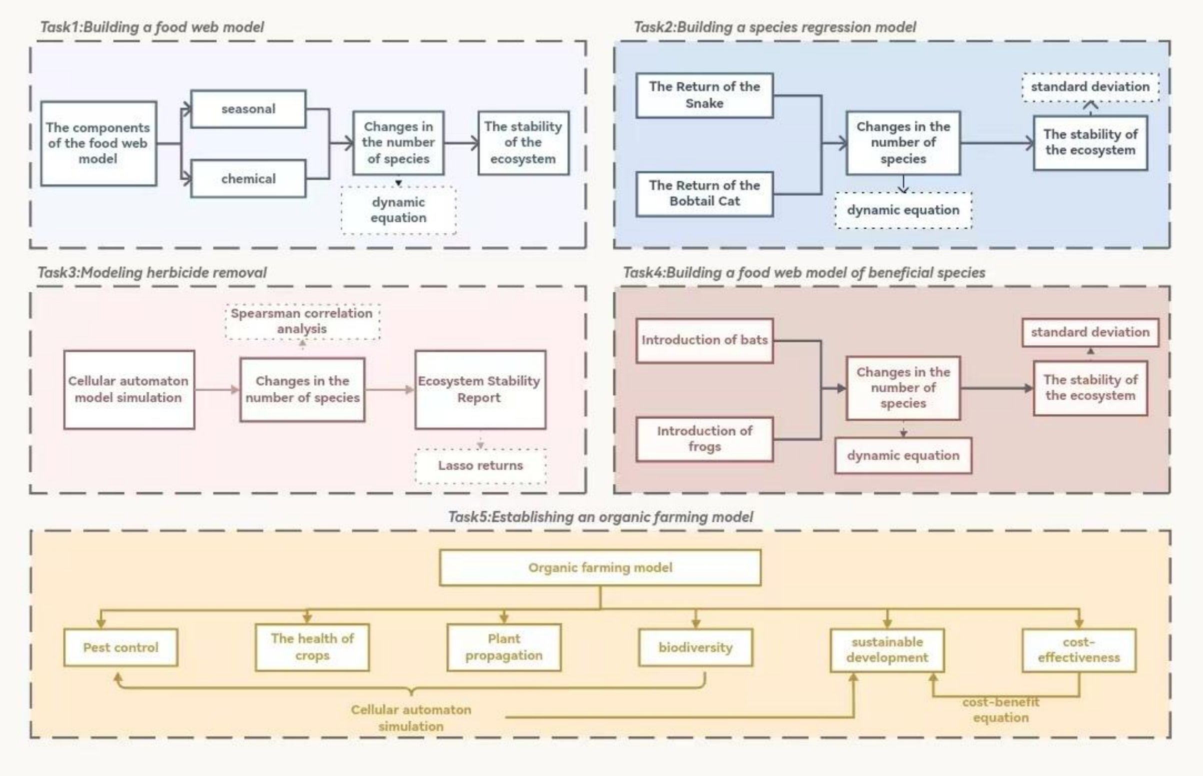

Figure 1: Our Work

2 Assumptions and Justifications

- Assumption1: The data we use is accurate and valid.
- ✓ Explanations: The data in this article comes directly from the latest results of online official databases and published literature.
- Assumption2: The forest is in a stable balance before being converted to farmland, and the number of species and ecological functions are stable
- ✓ Explanations: In nature, mature forest ecosystems have undergone long-term succession, forming complex and stable relationships among species, and energy flow and material cycle have also reached a relatively stable state.
- Assumption3: After farmers remove herbicides, ecosystems move towards a new equilibrium based on their ability to repair themselves.
- ✓ Explanations: After stopping the use of herbicides, weeds grow naturally, providing food and habitat for insects, attracting birds, and the soil microbial community gradually recovers, improving soil structure and nutrient cycling

3 Notations

The key mathematical notations used in this paper are listed in Table 1.

Table 1: Notations used in this paper

Symbol	Description	Unit
P(t)	The population size of producers (plants/crops)	-
I(t)	Population size of primary consumers (pests)	-
C(t)	Population size of secondary consumers (birds/bats)	-
B(t)	Snake population	-
S(t)	Bobtail cat population	-
$I_b(t)$	Population size of pests	-
$I_g(t)$	The population of beneficial insects	-
F(t)	Frog population	-

4 Agroecosystem food web model

In order to simulate the agroecosystem that has recently replaced the dense forest area, we use Lotka-Volterra model and write code in combination with MATLAB to establish an agroecosystem food web model to analyze the changes in this agroecosystem.

4.1 Components of a food web model

In a simplified agroecosystem (see Figure 1), these components are typically included:

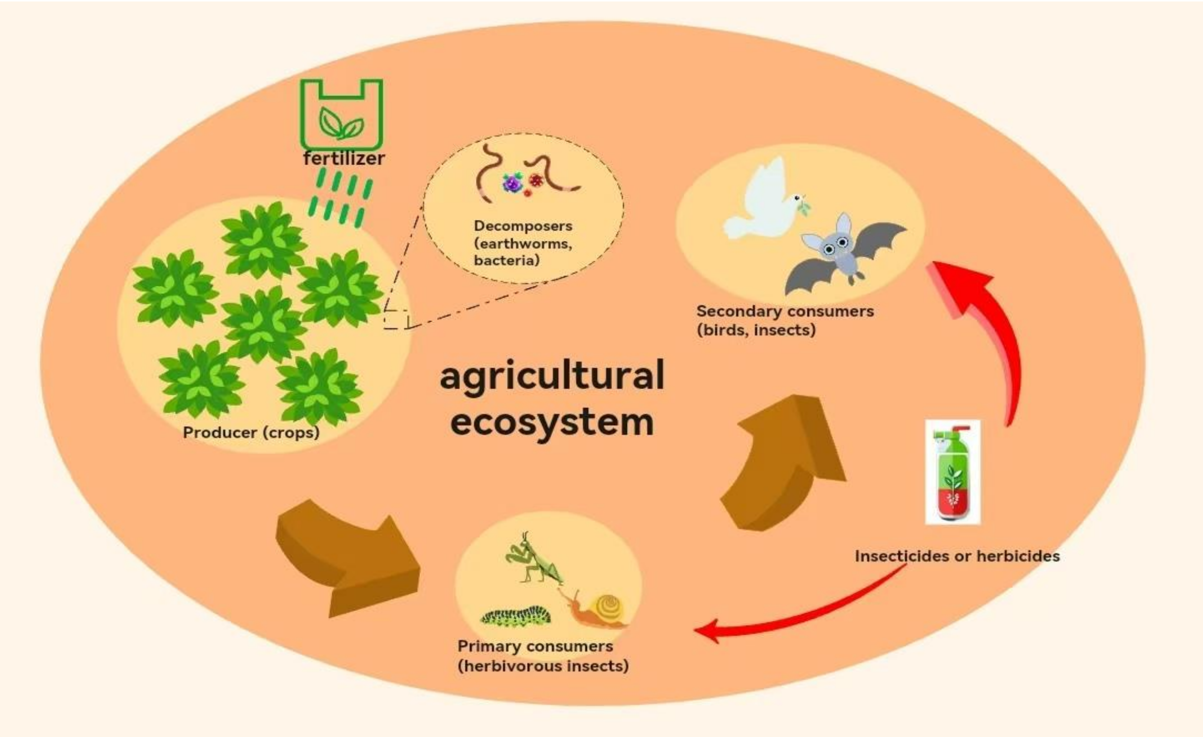

Figure 2: Agroecosystems

- Producers, i.e. plants, convert solar energy into chemical energy through photosynthesis, forming the foundation of agricultural ecosystems. With the continuous construction of farmland, the variety of plants has decreased significantly, from the original multi-type plants to a few crops.

- Primary consumers, that is, herbs, as farmland continues to be built, as the variety of plants decreases, the variety of herbs that feed on plants decreases, gradually evolving into crop-eating pests. Too many pests may cause crop yields to decrease, and pesticides can reduce the number of pests.
- Secondary consumers, i.e. birds and bats that feed on pests, their numbers vary with the number of pests, so the use of pesticides can also affect their numbers.
- Advanced consumers, such as leopards and short-tailed cats that feed on birds and bats, whose numbers vary with the number of birds and bat demons.
- Decomposers, i.e. microorganisms and earthworms in the soil, can promote the decomposition of organic matter, improve soil fertility, and benefit plant growth and development.

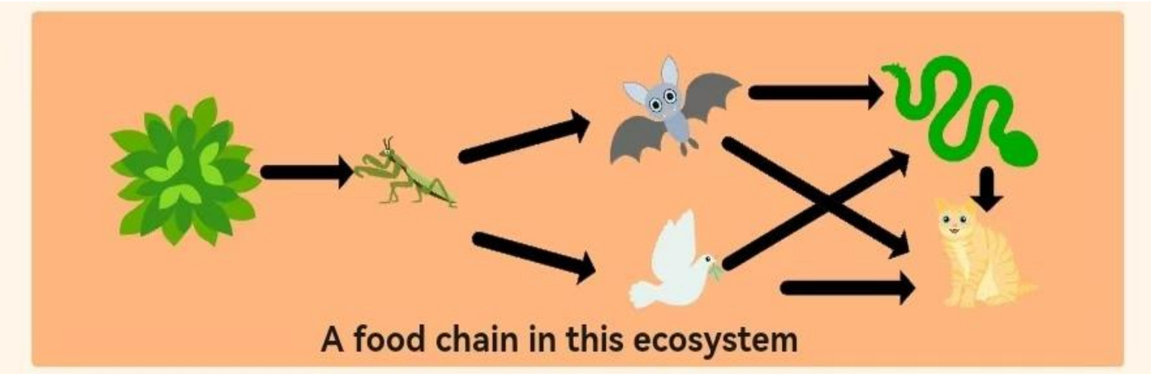

Figure 3: A food chain in an agricultural

These components make up one of the ecosystems as shown in Figure 2. In addition, in agricultural ecosystems, people also use chemicals to increase yields. On the one hand, people use pesticides to reduce the number of pests, but the abuse of pesticides will cause a decrease in the number of birds and bats, which in turn affects the number of short-tailed cats and snakes. It may cause a decrease in species diversity and reduce the stability of the ecosystem. On the other hand, people also apply chemical fertilizers to increase crop yields. Although chemical fertilizers can promote the growth and reproduction of decomposers in the short term, long-term application of chemical fertilizers may lead to problems such as soil pH change and land hardening, thereby inhibiting the growth and reproduction of decomposers and reducing the stability of the ecosystem.

4.2 Building a food web model

We can describe the seasonal dynamics of various populations by establishing differential equations [2], and then study the effects of chemical use on plant health, insect populations, bat and bird populations, and ecosystem stability.

First, we establish a seasonal forward coefficient m(t) to quantify the seasonal change of the ecosystem. With the change of seasons, abiotic factors such as light, temperature, and water in the ecosystem change (Figure 4).

Figure 4: Effect of seasonality on biological activity

We determine the value of the m (t) function by evaluating the impact of abiotic factors on different organisms. Where the ecosystem is more active in summer, the value of the m (t) function is set to 3, and the value of the ecosystem activity is reduced in winter. The value of the m (t) function is set to 1 to 3, and the spring and autumn are located between them. The value of the m (t) function is set to 1.

$$m(t) = \begin{cases} 1, t \in (4k, 4k+1) \cup (4k+2，4k+3) \\ \frac{1}{3}, \quad t \in (4k+3, 4k+4) \\ 3, \quad t \in (4k+1, 4k+2) \end{cases} \tag{1}$$

Where t is the month,$(4k, 4k+1)$: spring,$(4k+1, 4k+2)$:summer,$(4k+2，4k+3)$:autumn,$(4k+3, 4k+4)$:winter, we considered seasonality and combined the Lotka-volterra model to create the Se-LV model:
The differential equation for producers (plants/crops):

$$\frac{dP}{dt} = g_P P m(t)\left(1 - \frac{P}{K_P}\right) - \alpha_P PI + w_P P - \gamma_P P \tag{2}$$

In the equation,g_P:the natural growth rate of plants， K_P:the environmental carrying capacity of plants.α_P:the predation coefficient of pests on plants, ω_P: the promotion coefficient of fertilizers to plants， γ_P: the negative impact of herbicides on plants.
Differential equations for primary consumers (insects):

$$\frac{dI}{dt} = g_I I m(t)\left(1 - \frac{I}{K_I}\right) + \beta_P P - \frac{D_I I}{m(t)} - \gamma_I I - \alpha_I IC \tag{3}$$

In the equation,g_I: the natural growth rate of insects,K_I:the environmental carrying capacity of insects,β_P: the ability of pests to prey on plants, $D_I$indicate natural mortality of insects,γ_I: the killing coefficient of insects by insecticides,α_I: the secondary consumer predation coefficient for insects.
Differential equations for secondary consumers (birds and bats):

$$\frac{dC}{dt} = g_C C m(t)\left(1 - \frac{I}{K_C}\right) + F_I I - \frac{D_C C}{m(t)} - \gamma_C P \tag{4}$$

In the equation,g_C: the natural growth rate of secondary consumers.K_C:the environmental carrying capacity of secondary consumers,F_I: the ability of secondary consumers to prey on primary consumers, D_C: the natural mortality rate of birds and bats.γ_C: the killing coefficient of insecticides to birds and bats

4.3 Dynamic changes in each component

To study the effects of herbicides and insecticides on plant health, insect populations, bat and bird populations, and ecosystem stability [3], we used MATLAB to assist in mapping the number of producers, primary consumers (pests), secondary consumers (birds and bats) over time in the absence of herbicides (Figure 5), and the number of all three over time in the presence of herbicides (Figure 6).

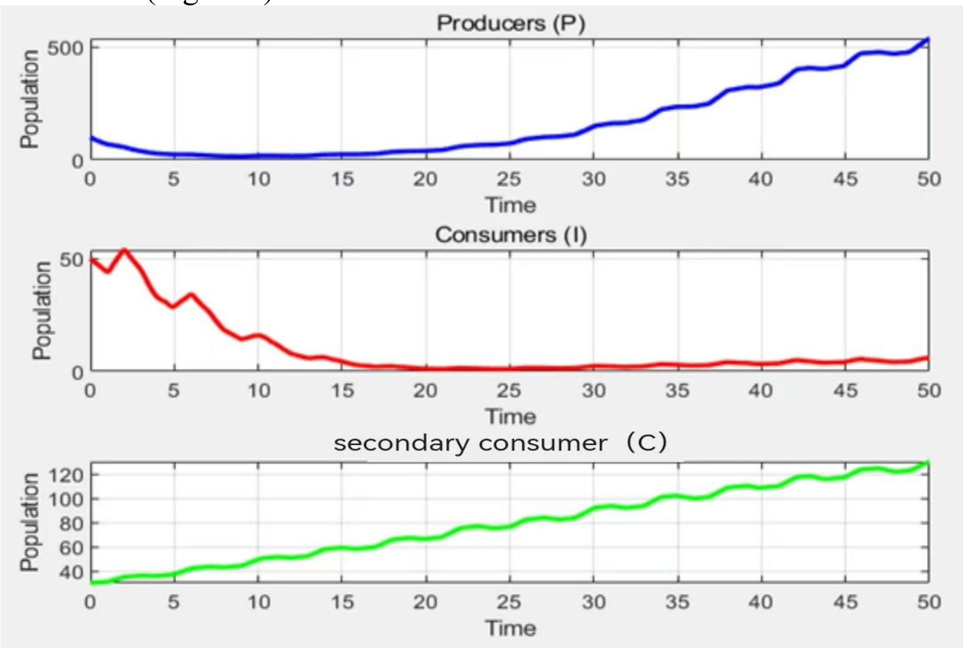

Figure 5: Changes in the number of components in the absence of herbicides

From Figure 5, it can be seen that the number of producers has continued to rise over time, gradually increasing from a low level at the outset to nearly 500.It shows that when herbicides are not used, the ecological environment is relatively natural, and crops can obtain sufficient sunlight, water, nutrients and other growth factors, which can stabilize growth and reproduction

The initial number of pests was around 50, with obvious fluctuations. It decreased first, then increased and then decreased, and then gradually decreased and stabilized at a lower level in a period of 15-50. It shows that the change of crop quantity directly affects the food supply of pests, and there are natural predation relationships and other factors in the ecosystem. The early number fluctuation may be due to the combined effect of changes in food resources and changes in the number of natural enemies (birds and bats, etc.). Later, under the natural regulation of the ecosystem, the number of pests reached a relatively stable level at a lower level.

The number of birds and bats continued to rise from a low level at the beginning, and approached 120 at the time of 50:00. Indicating that the ecosystem is rich in food resources such as pests, providing a suitable habitat for birds and bats to survive and breed

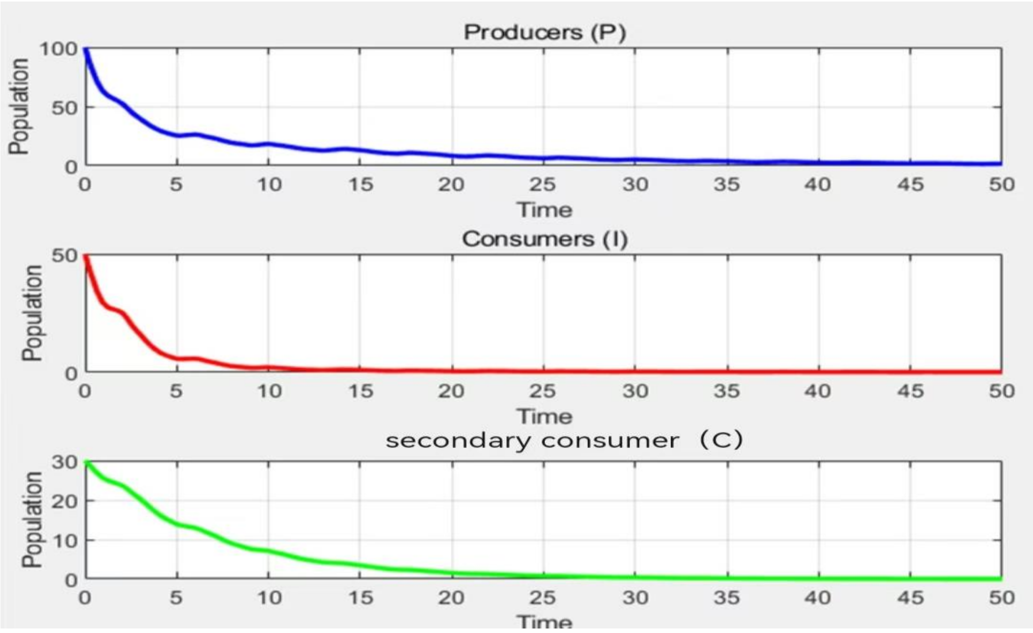

Figure 6 : Changes in the amount of each component in the presence of herbicides

Figure 6 shows that the initial number of producers (crops) is 100, and then it decreases sharply, stabilizing at a low level of nearly 20 over a period of 5-50. It indicates that after the use of herbicides, herbicides may have caused toxicity to crops, affecting the normal growth and physiological functions of crops, and inhibiting their reproduction

The initial number of pests is 50, and then it decreases rapidly, stabilizing at a very low level close to 0 in a period of 5-50. It indicates that herbicides may directly poison pests or destroy the living environment of pests. At the same time, the reduction of the number of crops makes the food resources of pests scarce, and a combination of factors leads to a sharp decrease in the number of pests.

Birds and bats start out at 30, then decline rapidly, stabilizing at very low levels near 0 over a period of 5-50. Due to the sharp decrease in pest populations, the main food sources for birds and bats have been reduced, and chemicals such as herbicides may also be toxic to birds and bats, affecting their survival and reproduction, resulting in a significant decrease in their numbers and maintaining extremely low levels.

Combined with Figures 5 and 6, it can be seen that the use of chemicals (herbicides) has a serious negative impact on ecosystem stability. It disrupts the food chain and food web structure between producers, consumers, and birds and bats, resulting in a sharp decrease in biomass and biodiversity.

5 Species Regression Models

As time went by, marginal habitats began to mature, and the region's native species of snakes and short-tailed cats gradually returned

5.1 The return of snake

Snakes feed on birds and bats in farmland ecosystems, and snakes are also prey targets for short-tailed cats. Considering snake hibernation, a seasonal function of snakes is established

$$\boldsymbol{\varphi}(\mathbf{t}) = \begin{cases} \mathbf{1}, \boldsymbol{t} \in (\mathbf{4}\boldsymbol{k}, \mathbf{4}\boldsymbol{k}+\mathbf{1}) \cup (\mathbf{4}\boldsymbol{k}+\mathbf{2},\ \mathbf{4}\boldsymbol{k}+\mathbf{3}) \\ \mathbf{0}, \quad \boldsymbol{t} \in (\mathbf{4}\boldsymbol{k}+\mathbf{3}, \mathbf{4}\boldsymbol{k}+\mathbf{4}) \\ \mathbf{3}, \quad \boldsymbol{t} \in (\mathbf{4}\boldsymbol{k}+\mathbf{1}, \mathbf{4}\boldsymbol{k}+\mathbf{2}) \end{cases} \tag{5}$$

From the habits of snakes, the differential equation of snakes can be established.

$$\frac{dS}{dt} = g_S S\varphi(t)\left(1-\frac{S}{K_S}\right) - \alpha_S SB \tag{6}$$

In the equation,$\boldsymbol{g_S}$indicates the natural growth rate of the snake.$\boldsymbol{K_S}$indicates the environmental carrying capacity of the snake,$\boldsymbol{\alpha_S}$represents the predation coefficient of short-tailed cats towards snakes

5.2 The return of short-tailed cat

In the farmland ecosystem, short-tailed cats not only prey on birds and bats, but also on snakes and have no obvious seasonal changes. From this, we can establish the differential equation of short-tailed cats:

The differential equation of the short-tailed cat:

$$\frac{dB}{dt} = g_B B\left(1-\frac{B}{K_B}\right) \tag{7}$$

In the equation,g_B: the natural growth rate of the bobtail cat,K_BIndicates the environmental carrying capacity of a short-tailed cat

5.3 Effects of species return

The return of species will inevitably affect the number of other species in the ecosystem, and will also affect the stability of the ecosystem and abiotic factors.

5.2.1 Changes in the number of species

Due to the introduction of natural enemies, the number of birds and bats will change significantly, and the differential equation for secondary consumers (birds and bats) is established again.

$$\frac{dC}{dt} = g_C Cm(t)\left(1-\frac{I}{K_C}\right) + F_I I - \frac{D_C C}{m(t)} - \gamma_C P - \alpha_C SC \tag{8}$$

In the equation,$\alpha_C$represents the predation coefficient of birds and bats by advanced consumers. There are dynamic changes in biomass across the food chain due to changes in bird and bat populations

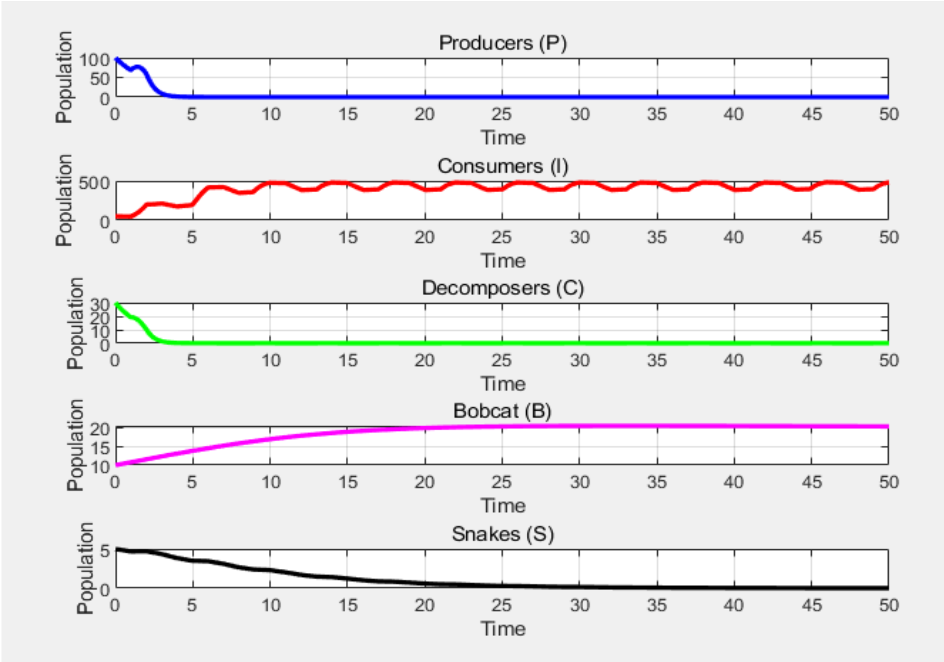

Figure7: Changes in the number of components in the absence of herbicides (after introduction)

Comparing Figures 7 and 5, it is found that the number of producers decreased initially, and then stabilized. After the return of snakes and short-tailed cats, short-tailed cats may prey on some consumers who feed on producers, which reduces the pressure of consumers on producers in the early stage, and the number of producers recovers and stabilizes to a certain extent.

The number of pests decreased significantly, and then fluctuated at a lower level. Snakes and short-tailed cats acted as predators, directly preying on consumers, resulting in a significant decrease in consumer populations, changing the species structure and quantitative relationships within consumers.

The number of birds and bats decreased in the early stage, but remained relatively stable after that. Snakes and short-tailed cats may prey on birds and bats, causing their numbers to decrease in the early stage. As the ecosystem adjusts, birds and bats adapt to new environments by changing their behavior (such as changing the time of activity, finding more hidden habitats), etc., and the number of snakes as prey objects of short-tailed cats gradually decreases, and the number of short-tailed cats as the top of the food chain gradually increases and then stabilizes. The ecosystem reaches a new dynamic balance under the new species relationship.

5.2.2 Stability of ecosystems

We introduce standard deviation to study the stability of ecosystems. Generally speaking, the smaller the standard deviation, the more stable it is. First, in order to eliminate the different effects of various community units, we perform Min-Max pretreatment on each population at different time. We pass the formula $std(X) = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \bar{x})^2}$,calculate the standard deviation of the number of species at different time points as shown in the figure

Figure 8: Ecosystem stability standard deviation

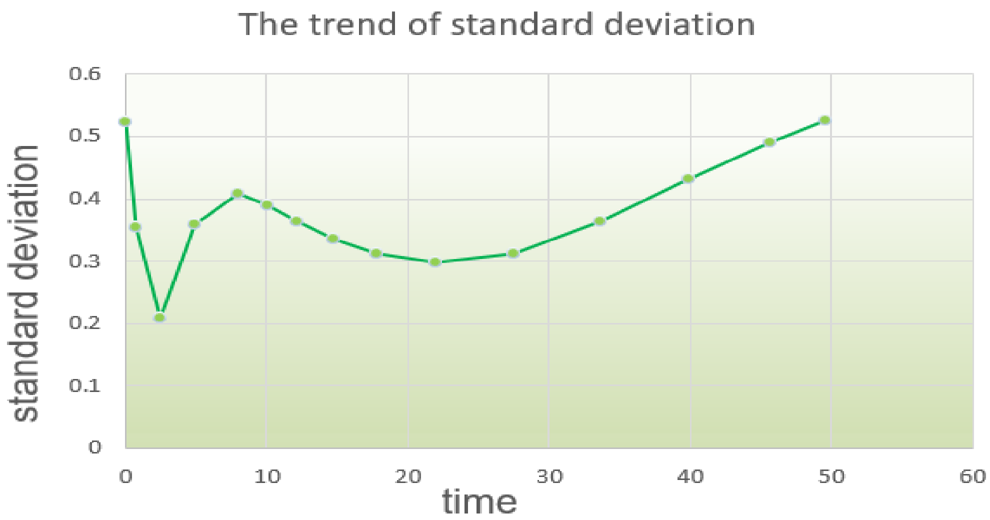

Figure 8: Ecosystem stability standard deviation

The low standard deviation of ecosystem stability suggests that the new balance may have enhanced the stability of the ecosystem. Snakes and short-tailed cats, as top predators, control the number of consumers, avoid the damage to producers caused by their overbreeding, and maintain the stability of ecosystem structure and function.

6 Herbicide removal models

We can create a cellular automaton model with the help of MATLAB, which visually shows the survival status of each population in the farmland with or without herbicides in a graphical way. In addition, using MATLAB to calculate the proportion of each population from 1 to 60 generations in a unit of 5 steps. Then spearman correlation analysis is done on each set of data, and finally lasso regression is used to determine the relationship between various groups to report the stability of the ecosystem.

6.1 Cellular Automata Model

Cells only interact with their neighbors. This local interaction principle is similar to the way organisms interact with each other and with the environment in an ecosystem. Therefore, cellular automata can be used to study ecosystem iterations with or without herbicides. From biological knowledge, we can establish the following cellular rules:

- Rule 1: If there are 5 or more open spaces or weeds around a normal individual and more than three generations, it will enter a weakened state, and if there are 5 or more open spaces or weeds around a weakened individual and more than three generations, it will die

  Explanation: Individuals are exposed to a relatively scarce resource or competitive pressure for a long time, which affects their physiological functions
- Rule 2: After death, plots of land are randomly converted to open fields or weeds or crops with equal probability

  Explanation: Simulates multiple possibilities in the natural environment

- Rule 3: If you step five units in time, the individual reproduces once, randomly generating two animal plots around it
  Explanation: Simulates the cyclical reproduction of organisms
- Rule 4: With each step in time, the individual will move one square in a random direction, and the previous plot will randomly become vacant land or weeds or crops.
  Explanation: Models the random search behavior of organisms or random diffusion processes influenced by environmental factors
- Rule 5: Crops become weeds when they are surrounded by 5 or more weeds and persist for more than three generations, and weeds become crops when they are surrounded by 5 or more weeds and persist for more than three generations
  Explanation: Communities with a comparative advantage have stronger allelopathy
- Rule 6: A clearing has a 5% chance of becoming a weed
  Explanation: Simulates the natural process of weed seeds germination and growth in a clearing
- Rule 7: With herbicides, crops have a 5% chance of being cleared (simulated killing by herbicides) and weeds have a 10% chance of being cleared (simulated killing by herbicides). Without herbicides, crops have a 2.5% chance of being cleared and weeds have a 2.5% chance of being cleared
  Explanation: Herbicides interfere with important physiological processes such as photosynthesis, respiration, or cell division in plants, and have specific herbicidal activities

According to the above rules, We can use MATLAB to map the cellular automaton with or without herbicides (Figure 9).

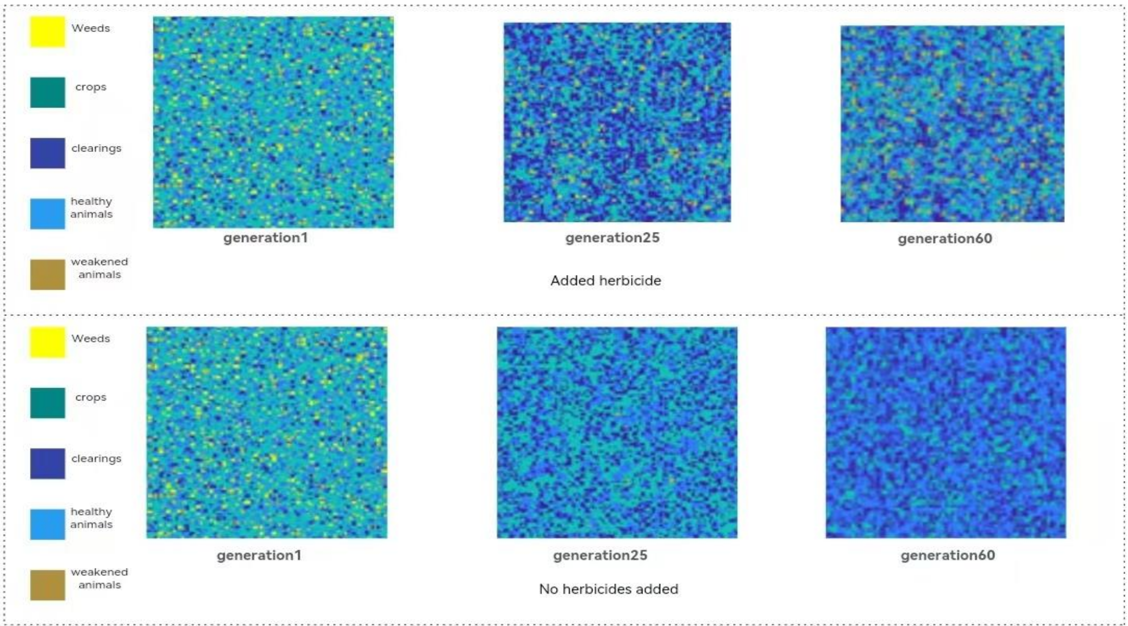

Figure 9: Cellular automaton image magnified by a certain number

In order to quantify the quantitative changes of various populations with or without herbicides, the proportion of each population from 1 to 60 generations was counted
The situation and the normal analysis of the data found that it does not conform to the normal distribution, and considering that they are all quantitative variables, the spearman correlation analysis was used to obtain the degree of correlation heat table (Table 2, 3).

Table 2: Herbicide removal models

Table 3: Correlation of various variables in pesticide use

6.2 Ecosystem stability report

Producer growth and numbers are influenced by environmental factors such as light, moisture, and soil nutrients, as well as interactions with other organisms. Studying the impact of herbicide removal on producers involves complex variable correlations. Lasso regression can screen for variables that play a key role in producer stability, simplify models, avoid overfitting, and help understand the mechanisms of change in ecosystem stability at the producer level.

Consumers are divided into multiple levels, and their predation relationships are complex, and survival and population changes are affected by various factors such as predation relationships, food resource abundance, and living space. Analyzing the impact of herbicide removal on consumers, the data dimension is high and the variables have a large mutual influence. Lasso regression effectively handles the multicollinearity problem among variables by imposing L1 norm constraints on the coefficients, making the model estimation more stable and reliable, and then reveals the changing law of ecosystem stability at the consumer level.

So we chose Lasso regression to report the stability of the ecosystem. We studied the lasso regression with crops as the independent variable and other factors in the table as the dependent variable in the two periods with or without herbicides, and the lasso regression with healthy animals as the dependent variable and other factors in the table as the independent variable. From the results of the study, we can draw the following conclusions:

From the producer's perspective, herbicides inhibit weed growth, reduce the competition between weeds and crops, and benefit crop growth.From a consumer perspective, herbicides may affect animal health through the food chain, altering the proportion of healthy and frail animals, which in turn affects ecosystem balance.Overall, the use of herbicides may benefit crop growth in the short term, but may have potential negative impacts in terms of long-term ecosystem stability and biodiversity.

7 Beneficial species food web model

7.1 Introduction of bats

Assuming that the farmland ecosystem is not using herbicides at this time, we simulate bats as insectivores that control pest populations and pollinators that support plant reproduction [4], and the differential equation of change for producers is:

$$\frac{dP}{dt} = g_P P m(t)\left(1-\frac{P}{K_P}\right) - \alpha_P P I_b + w_P P + C_{bP} B \tag{9}$$

C_{bP}indicates the bat's help to plants (supports plant reproduction)

Differential equation for pests:

$$\frac{dI_b}{dt} = g_{I_b} I_b m(t)\left(1-\frac{I_b}{K_{I_b}}-\frac{\alpha_{g_b} I_g}{K_{I_b}}\right) - \frac{D_{I_b} I_b}{m(t)} - \alpha_{I_b} I_b B - \alpha_{I_b} I_b C \tag{10}$$

In the equation,g_{I_b}indicates the natural growth rate of pests.K_{I_b}indicates the environmental carrying capacity of a pest. D_{I_b}indicates the natural mortality rate of pests, α_{I_b}represents the secondary consumer predation coefficient for pests.

Differential equation of beneficial insects:

$$\frac{dI_g}{dt} = g_{I_g} I_g m(t)\left(1-\frac{I_g}{K_{I_g}}-\frac{\alpha_{b_g} I_b}{K_{I_g}}\right) - \frac{D_{I_g} I_g}{m(t)} - \alpha_{\mathrm{I}} I_g C \tag{11}$$

In the equation,g_{I_g}indicates the natural growth rate of beneficial insects,K_{I_g}indicates the environmental carrying capacity of beneficial insects, D_{I_g}Indicates the natural mortality rate of beneficial insects, α_{I_g}Indicates the predation coefficient of secondary consumers towards beneficial insects

Differential equation for secondary consumers:

$$\frac{dC}{dt} = g_C Cm(t)\left(1-\frac{C}{K_C}\right)-\frac{D_C C}{m(t)}-\alpha_C CS \tag{12}$$

The differential equation for bats:

$$\frac{dB}{dt} = g_B Bm(t)\left(1-\frac{B}{K_B}\right)-\alpha_B BS+\alpha_{I_b} I_b B-\frac{D_B B}{m(t)} \tag{13}$$

In the equation,$g_B$indicates the natural growth rate of bats,$K_B$indicates the environmental carrying capacity of bats, $D_B$indicates the natural mortality rate of bats

Variable differential equation of snakes:

$$\frac{dS}{dt} = g_S S\varphi(t)\left(1-\frac{S}{K_S}\right) \tag{14}$$

7.2 Introduction of frogs

The differential equation of the frog:

$$\frac{dF}{dt} = g_F F\left(1-\frac{F}{K_F}\right)+\alpha_F I_B-\alpha_{SF} SF \tag{15}$$

In the equation,$g_F$indicates the natural growth rate of the frog,$K_F$indicates the environmental carrying capacity of the frog,$\alpha_F$indicates the frog′s ability to catch pests,α_{SF}represents the ability of snakes to prey on frogs. Since frogs have an evolutionary role in their environment, we define the environmental impact of frogs as benefit F =k_FF, $k_F$is the purification factor. After the introduction of frogs, the number of components in the ecosystem changed

The differential equation of change for producers is:

$$\frac{dP}{dt} = g_P Pm(t)\left(1-\frac{P}{K_P}\right)-\alpha_P PI_b+w_P P+C_{bP} B+\text{benefitF} \tag{16}$$

Differential equation for pests:

$$\frac{dI_b}{dt} = g_{I_b} I_b m(t)\left(1-\frac{I_b}{K_{I_b}}-\frac{\alpha_{g_b} I_g}{K_{I_b}}\right)-\frac{D_{I_b} I_b}{m(t)}-\alpha_{I_b} I_b B-\alpha_{I_b} I_b C+\text{benefitF} \tag{17}$$

Differential equation of beneficial insects:

$$\frac{dI_g}{dt} = g_{I_g} I_g m(t)\left(1-\frac{I_g}{K_{I_g}}-\frac{\alpha_{b_g} I_b}{K_{I_g}}\right)-\frac{D_{I_g} I_g}{m(t)}-\alpha_{\mathrm{I}} I_g C+\text{benefitF} \tag{18}$$

Differential equation for secondary consumers:

$$\frac{dC}{dt} = g_C Cm(t)\left(1-\frac{C}{K_C}\right)-\frac{D_C C}{m(t)}-\alpha_C CS+\text{benefitF} \tag{19}$$

The differential equation for bats:

$$\frac{dB}{dt} = g_B Bm(t)\left(1-\frac{B}{K_B}\right)-\alpha_B BS+\alpha_{I_b} I_b B-\frac{D_B B}{m(t)}+\text{benefitF} \tag{20}$$

Variable differential equation of snakes

$$\frac{dS}{dt} = g_S S\varphi(t)\left(1-\frac{S}{K_S}\right)+\text{benefitF} \tag{21}$$

7.3 Comparison of beneficial species

7.3.1 Number of species

We again used MATLAB to map the changes in the number of species before and after the introduction of frogs (Figure 10).

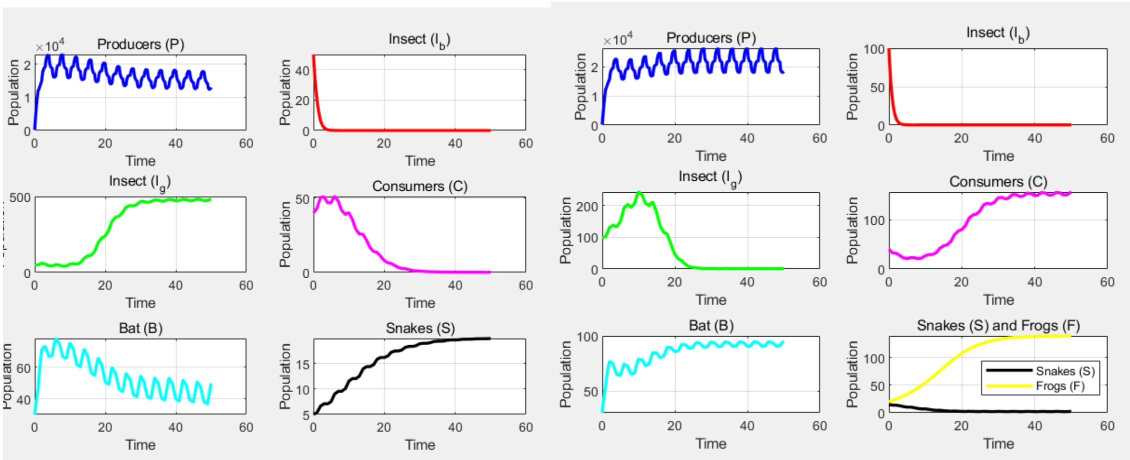

Figure 10 ： Changes in the number of species before and after the introduction of frogs

As can be seen from the figure, both bats and frogs have a certain control effect on insects. Bats make the number of insects fluctuate and maintain within a certain range; frogs have a significant effect on inhibiting the growth of insects and reducing them to lower levels. Because frogs themselves are consumers in the ecosystem, their presence increases the biodiversity of the ecosystem and the complexity of the food chain. Frogs provide more energy sources for predators such as snakes, thus promoting the growth of snake populations. At the same time, the control of pests by frogs allows plants in the farmland ecosystem to grow better, providing more food resources for other herbivorous consumers, which in turn promotes the increase of the consumption population.

7.3.2 Stability of ecosystems

We use the same method as 5.3.2 to calculate the standard deviation of the number of individual species in the ecosystem at this time

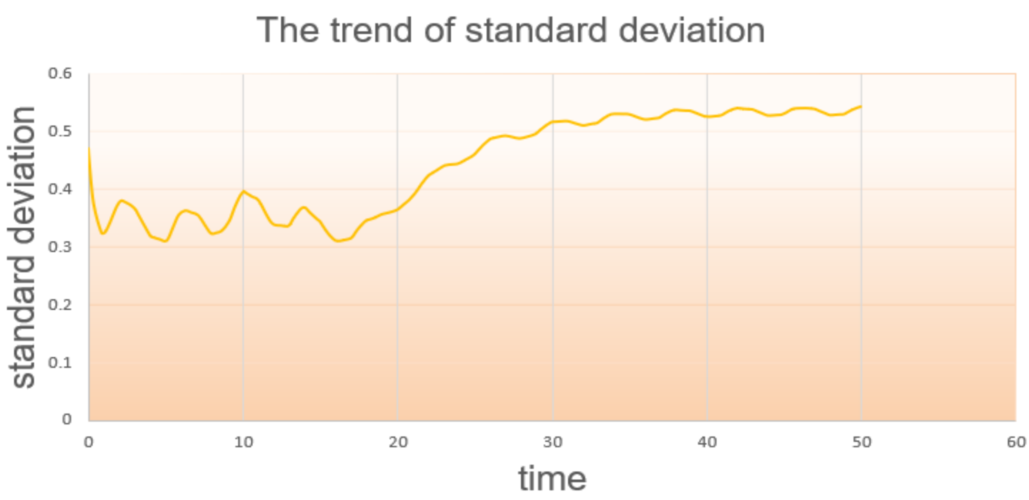

Figure 11: Standard deviation of the ecosystem

The standard deviation of ecosystem stability was lower than before, indicating that the introduction of frogs enhanced ecosystem stability.

8 Organic farming models

We use cellular automata models and dynamic equations to discuss the impact of organic farming on pest control, crop health, plant reproduction, biodiversity, long-term sustainability, and cost-effectiveness.

8.1 The concept of organic farming

Organic farming is an agricultural production method based on ecological principles and the concept of sustainable development. It emphasizes the minimization or avoidance of the use of synthetic chemicals, such as fertilizers, pesticides, and growth regulators, in the agricultural production process, and instead makes full use of the functions of natural ecosystems and the interrelationships between organisms to achieve the goals of agricultural production.

8.2 Cellular Automata Model

In order to explore the impact of farmers' adoption of organic farming practices on farmland ecosystems, we established cellular automata to simulate the impact of organic farming on agroecosystems. By dynamically simulating grid state changes, the following key indicators can be analyzed: trends in healthy crops, dynamic changes in pest populations, changes in biodiversity over time, and cost-effectiveness of farming methods.

First we define four 50x50 2D grids, where each cell represents a small plot. Then we set the update rules of the cellular automaton as follows

Rule 1: Random distribution of initial soil fertility and state

Rule 2: Organic farming can improve soil fertility, increase biodiversity, and reduce pest density

Rule 3: The spread of pests from high-density areas to neighboring units is negatively correlated with biodiversity.

Rule 4: Crop health is influenced by soil fertility and pest density, and healthy crops are more likely to multiply and expand to neighboring units.

Rule 5: Organic farming increases biodiversity and promotes the growth of natural enemies of pests.

Rule 6:

Set the different states in which four categories of factors may be present:

Crop health status (1 = healthy, 0 = unhealthy, -1 = pest infestation)

Soil fertility (1 = high, 0 = medium, -1 = low)

Pest density (1 = none, 0 = low, -1 = high)

Biodiversity (1 = high, 0 = medium, -1 = low)

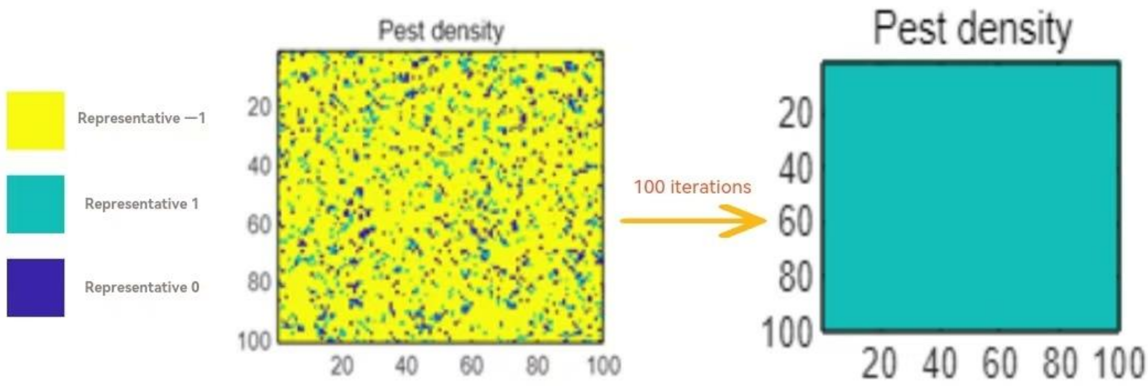

Figure 12: Changes in pest density

As can be seen from Figure 12, the density of pests decreases during the process of organic farming. In practice, organic farming mainly relies on natural pest control mechanisms. For example, by releasing the natural enemies of pests, insects, and using the predation and parasitic relationship between organisms to control the number of pests; using the sex pheromones of pests to trap pests and interfere with their mating and reproduction; using crop rotation, intercropping and other planting methods to disrupt the life history and habitat of pests, reduce the breeding and spread of pests, thereby maintaining the relative balance of pests and other organisms in the farmland ecosystem.

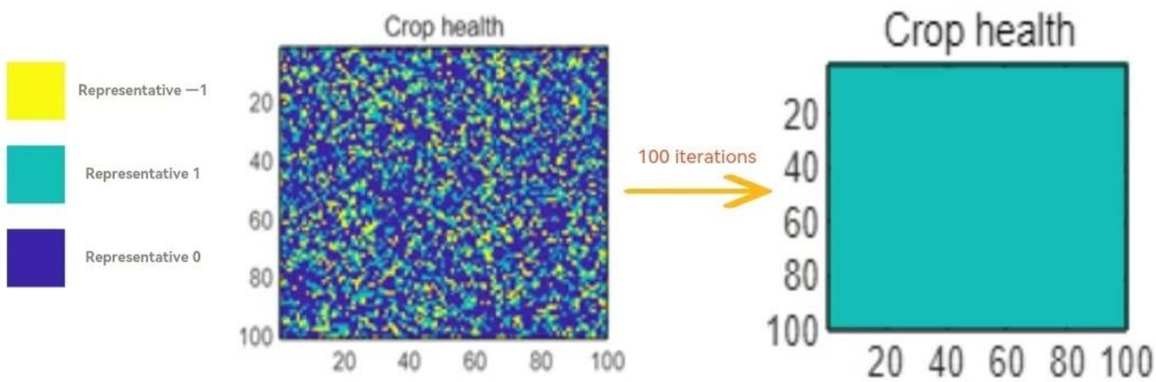

Figure 13: Changes in crop health status

It can be seen from Figure 7 that the health status of crops continues to improve during the process of organic farming. In practice, organic farming focuses on ensuring crop health by cultivating healthy soils. On the one hand, a large amount of organic fertilizers, such as compost, green manure, animal manure, etc., are applied to increase soil organic matter content, improve soil structure, improve soil water and fertilizer retention capacity and aeration, and create good conditions for crop root growth. On the other hand, a reasonable rotation system and intercropping mode are adopted to avoid continuous cropping obstacles, make full use of the complementary and mutually beneficial relationship between different crops, enhance crop immunity and stress resistance, so that crops can grow in a healthy environment and reduce the occurrence of diseases and pests.

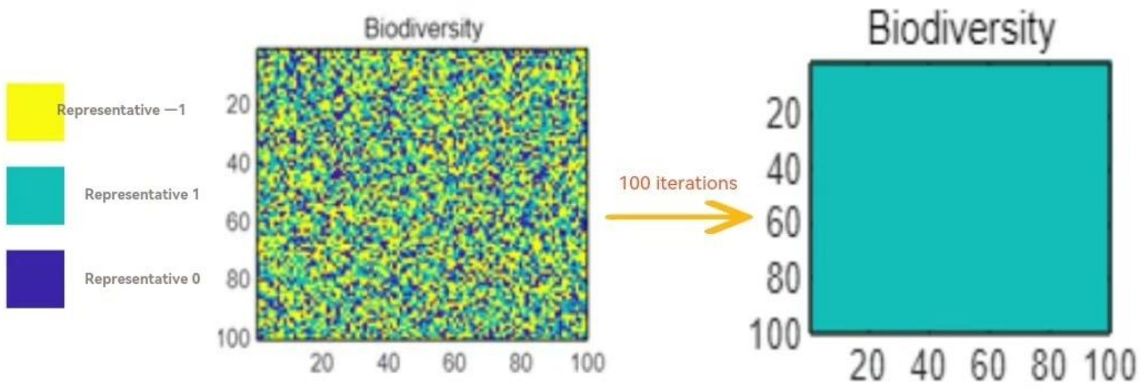

Figure 14: Changes in biodiversity

It can be seen from Fig. 14 that the biodiversity becomes richer in the process of organic farming. In practice, organic farmland does not carry out large-scale single crop cultivation, but adopts diverse planting patterns, including intercropping, intercropping, mixed cropping, etc., which provide rich habitat and food resources for different plants, animals and microorganisms. In addition, preserve the natural vegetation and ecological corridors around the farmland, reduce the disturbance and damage to the farmland ecosystem, promote the exchange and migration of species, make the farmland an ecosystem where multiple organisms coexist, and improve the stability and service functions of the entire ecosystem.

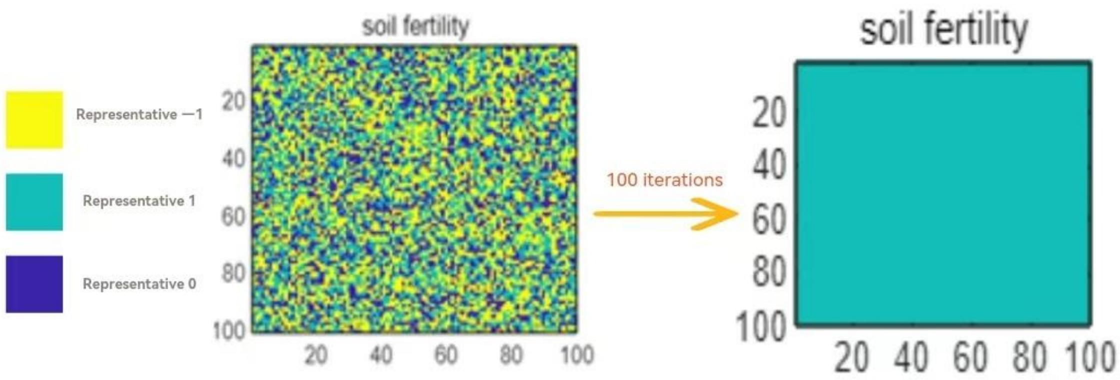

Figure 15: Changes in soil fertility

As can be seen from Figure 9, soil fertility is enhanced during the process of organic farming [5]. In practice, organic farming often applies compost, green manure and decomposed animal manure to provide nutrients for the soil and improve soil structure; implements the rotation of beans and grasses, and the intercropping of corn and soybeans to balance the use of nutrients and improve land utilization; inoculates rhizobium and other microbial agents to protect soil animals such as earthworms and enhance soil activity.

8.3 Long-term sustainability and cost-effectiveness

The dynamic relationship between measures taken by farmers, biodiversity (D), and soil fertility is as follows:

$$D = D_0 \cdot e^{-K_{F_D} C_{F_O}} \cdot e^{K_{M_D} M} \tag{22}$$

$$C_F = C_{F_0} \cdot e^{-K_{D_F}D} \cdot e^{-K_{M_F}M} \tag{23}$$

$$R_P = R_{P_0}\left(1 + K_D D^{\alpha} + K_S S^{\gamma} + K_M M^{\beta}\right) \tag{24}$$

$$C_P = C_{P_0}\left(1 + K_{M_C}M + K_{F_C}C_F\right) \tag{25}$$

From the above dynamic equation, it can be seen that organic farming can benefit the development of ecosystems in terms of improving biodiversity and increasing soil fertility, and is conducive to maintaining the stability and balance of ecosystems. However, at the same time, due to its high-tech content, the high cost of organic farming cannot be ignored. Therefore, we use the idea of linear programming to introduce the following cost-benefit formula to calculate the best ratio of traditional agricultural methods and organic agriculture.

$$Profit = R_P \cdot P - C_P \cdot P - C_F \cdot F - - \text{EcoImpact} \tag{26}$$

In the equation,EcoImpact = $\frac{e^{C_F}}{10}$ is an environmental damage factor, $R_P$is the income per unit of farmland. $C_P$is cost per unit of farmland.$C_F$is the amount of fertilizer used. Finally, through calculation, we get that when the proportion of organic agriculture is 0.63, the benefit value of agricultural production will reach the maximum.

9 Sensitivity Analysis

Next, we perform sensitivity analysis on the food chain model. Taking the increase of insect predation rate on plants as an example, we get the dynamic changes between the components in the following figure

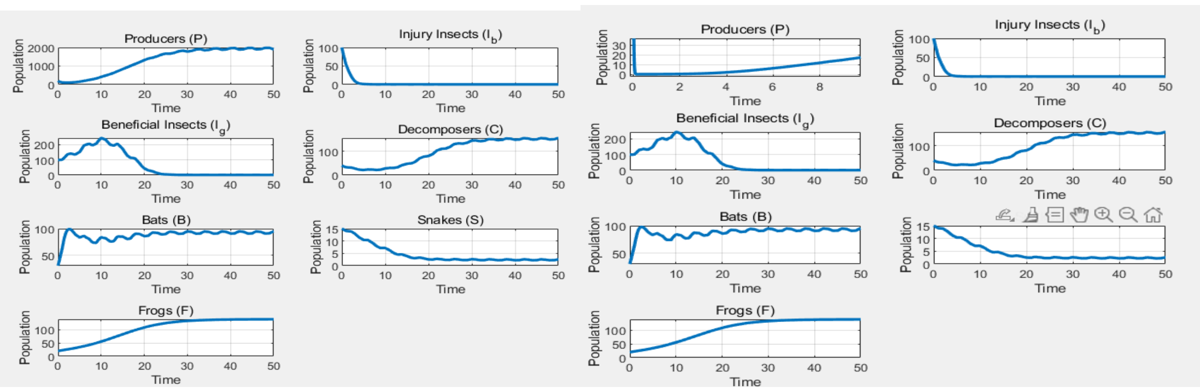

Figure 16: Before and after the increase in the predation rate of insects on plants

Combining the images, we can see that the natural growth rate of insects increases, and most of the data changes are more stable than the micro-novel model. At the same time, we change the natural growth rate of secondary consumers, the coefficient of secondary consumers preying on insects, the coefficient of secondary consumers preying on prey, the natural growth rate of bats, the natural growth rate of frogs, the natural growth rate of snakes, and the coefficient of snakes preying on frogs. Similar conclusions are obtained. When changing the environmental purification coefficient of frogs, we find that all images have great changes in trends and fluctuations, indicating that great caution is required when using this parameter for forecasting or decision-making.

10 Model Evaluation and Further Discussion

10.1 Strengths

- ✓ Dynamic equation model: can intuitively change the quantity of each component in the farmland ecosystem over time
  Through the structure and parameters of the equation, the mechanism and law of action of each component in the ecosystem can be revealed.
- ✓ Cellular automaton model: Starting from the cell and its rules at the microscopic level, complex phenomena and laws at the macroscopic level can naturally emerge, which can simulate the interaction between various components in the farmland ecosystem well, and visualize the impact of different changes on the entire ecosystem. And the update rules of the cell are relatively independent, suitable for parallel computing, which can improve computing efficiency, and can quickly obtain simulation results using parallel computing resources.
- ✓ Lasso regression model: It can automatically select variables in the estimation process, shrink some unimportant variable coefficients to 0, so as to filter out variables that have an important impact on the response variables, simplify the model structure, and improve the interpretability of the model. At the same time, the L1 regularization penalty is used to prevent overfitting, and the calculation efficiency is high.

10.2 Weaknesses

The dynamic equation model is complex and difficult to solve, and the numerical solution may have problems of stability and accuracy, and the calculation cost is high

There is a boundary in the cell space, and the neighbors of the cell at the boundary are different from those in the interior, which may lead to abnormal behavior at the boundary and affect the overall accuracy and authenticity of the simulation

10.3 Further Discussion

The dynamic equation model can be coupled with other models: coupled with other types of models such as discrete event models, statistical models, etc., to integrate the advantages of different models. Improve the accuracy and stability of numerical solutions

Boundary processing methods can be improved: periodic boundary conditions, reflective boundary conditions or dynamically expanding boundaries are used to reduce the influence of boundary effects on simulation results and make the system behave more naturally and reasonably at the boundary

11 Conclusion

By building models of food webs, cost-benefit models of agricultural production, etc., we have learned that in an ecosystem, organisms in every link of the food web play an indispensable role, and their symbiotic and antagonistic relationships are necessary to maintain the delicate system of nature. From forests to farms, humans are constantly transforming nature for survival, and various measures have been applied to agricultural production, such as the use of chemicals and the introduction of beneficial organisms. But as our research results show, the model is highly sensitive to all parameters, which means that our actions can easily lead to ecosystem disorders. Therefore, protecting the environment and maintaining the integrity of the food web must be embedded in the important ideas of production and life. Every beneficial means of production must be carefully studied to ensure its greenness. Sustainability actually includes not only environmental protection, but also economic benefits. Therefore, when making decisions, it is necessary to quantify costs and benefits to ensure profitability. Only by achieving the best in both green ecology and economy can a truly sustainable agricultural production method be achieved.

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A letter to farmers who are exploring organic farming methods

Dear farmer friends,

Hello!

It is very meaningful to know that you are exploring organic farming methods. It not only protects the environment and improves soil fertility, but also produces healthy and safe agricultural products and by-products such as meat, eggs and milk, which not only meets consumer demand for green food, but also generates higher economic income. Here are some suggestions, I hope they can help you.

Organic farming may face some economic challenges in the early stage. For example, the cost of organic fertilizers is relatively high, and the research and development and application of biological control technologies also require more manpower and material resources. However, in the long run, the benefits brought by organic farming are significant. First, organic agricultural products can usually be sold at higher prices and can increase farmers' income. Secondly, organic farming reduces the use of chemical drugs, helps to maintain ecological balance and achieve sustainable agricultural development. In addition, organic farming can also improve soil fertility and water retention capacity, avoid soil salinization, and reduce the impact of natural disasters on fields.

With regard to possible strategies, you can start by reducing the use of herbicides or pesticides and gradually shift to organic farming. This can let you gradually adapt to the management of organic farming. Secondly, you can also use organic fertilizers to improve soil structure, increase the content of organic matter in the soil, and improve soil fertility. At the same time, the flexible application of biological control technology can not only effectively control pests and diseases, but also reduce the dependence on chemical pesticides.

In terms of finding policy incentives, you can first seek government subsidies. The government can share the transformation risk of organic farming with farmers through agricultural aids. Such as subsidies given for the purchase of organic fertilizers, biological control equipment, etc.. You can also seek experts and technicians to help you master the technical points of organic farming and improve the level of farmland production management. At the same time, you can also seek the organic agricultural product certification system and market access mechanism established by the government to facilitate the sale of organic agricultural products. This can greatly increase your profits.

In short, organic farming is a cause that benefits the present and the future. Although there may be some difficulties in the early stage, as long as you have firm beliefs and actively take effective measures, you will be able to overcome difficulties and achieve sustainable agricultural development. I hope these suggestions are helpful to you. I wish you more and more progress on the road of organic farming and fruitful results!

Yours Sincerely